RoboBrain: A Unified Brain Model for Robotic Manipulation from Abstract to Concrete

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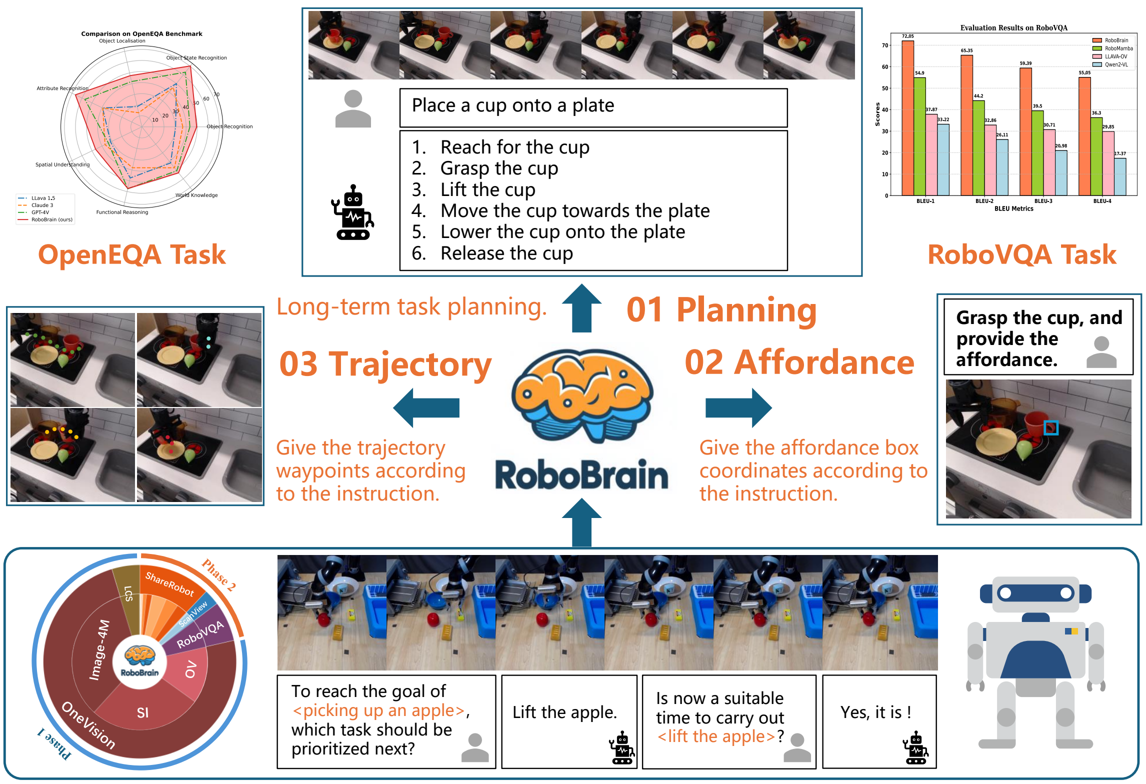

Figure 1. **Overview of RoboBrain.** RoboBrain consists of three key robotic capabilities: planning capability, affordance perception, and trajectory prediction. RoboBrain outperforms previous MLLMs in robotics tasks. The bottom part shows the composition of RoboBrain's training data and provides a specific example of visual question answering from our proposed ShareRobot. Best viewed on screen.

Abstract

Recent advancements in Multimodal Large Language Models (MLLMs) have shown remarkable capabilities across various multimodal contexts. However, their application in robotic scenarios, particularly for long-horizon manipulation tasks, reveals significant limitations. These limitations arise from the current MLLMs lacking three essential robotic brain capabilities: ***Planning Capability****, which involves decomposing complex manipulation instructions into manageable sub-tasks;* ***Affordance Per-***

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***ception**, the ability to recognize and interpret the affordances of interactive objects; and **Trajectory Prediction**, the foresight to anticipate the complete manipulation trajectory necessary for successful execution. To enhance the robotic brain's core capabilities from abstract to concrete, we introduce **ShareRobot**, a high-quality heterogeneous dataset that labels multi-dimensional information such as task planning, object affordance, and end-effector trajectory. ShareRobot's diversity and accuracy have been meticulously refined by three human annotators. Building on this dataset, we developed **RoboBrain**, an MLLM-based model that combines robotic and general multi-modal data, utilizes a multi-stage training strategy, and incorporates long videos and high-resolution images to improve its robotic manipulation capabilities. Extensive experiments demonstrate that RoboBrain achieves state-of-the-art performance across various robotic tasks, highlighting its potential to advance robotic brain capabilities. Project website: RoboBrain.*

1. Introduction

Recent advancements in Multimodal Large Language Models (MLLMs) have significantly advanced the pursuit of Artificial General Intelligence (AGI). By leveraging extensive multimodal datasets sourced from the internet and employing self-supervised learning techniques, MLLMs demonstrate exceptional capabilities in visual perception and understanding human language instructions, excelling in tasks such as visual question answering [3, 12, 13], image captioning [22, 29, 32], and sentiment analysis [14, 17]. Despite significant progress in MLLMs, the exploration of their application in robotics remains in its early stages, highlighting a crucial area for further research and innovation.

Recent studies have examined the application of MLLMs in robotics, focusing on planning and subgoal decomposition [6, 24], action sequencing [8, 9], and replanning and feedback [35, 38, 62]. However, their effectiveness in robotic scenarios—particularly for long-horizon manipulation tasks—reveals significant limitations. These limitations stem from the current MLLMs' lack of three critical robotic capabilities: planning, affordance perception, and trajectory prediction, as illustrated in Fig. 1. For instance, consider a robotic arm tasked with lifting a teapot and pouring water into a cup. The MLLM should be capable of decomposing this task into sub-tasks, such as "approach the teapot and lift it", "move the teapot until the spout is positioned over the cup", and "tilt the teapot to pour". For each sub-task, such as "approach and grasp the teapot", the MLLM must utilize affordance perception to accurately identify the graspable regions of the teapot. Additionally, trajectory prediction is essential for determining the complete path from the starting point to the graspable part of the teapot. This challenge for existing MLLMs primarily arises from the scarcity of large-scale, fine-grained datasets specifically designed for robotic operation tasks.

To empower the RoboBrain's core capabilities that transition from abstract instruction comprehension to concrete action expression. we first introduce ***ShareRobot***, a large-scale, fine-grained dataset specifically designed for robotic operation tasks. Specifically, we label multi-dimensional information such as task planning, object affordance, and end-effector trajectory. Building upon ShareRobot, we developed ***RoboBrain***, an MLLM model based on the LLaVA [34] architecture, aimed at enhancing the perception and planning capabilities of robots in complex tasks. In the process of training RoboBrain, we meticulously designed the ratio of robotic data to general multi-modal data, implemented a multi-stage training strategy, and incorporated long videos and high-resolution images. This approach endowed RoboBrain with powerful visual information perception capabilities in robotic scenarios, supporting historical frame memory and high-definition image input, thereby further enhancing the ability in robotic manipulation planning. Extensive experimental results demonstrate that RoboBrain outperforms existing models across multiple robotic benchmarks, including RoboVQA [50] and OpenEQA [42], achieving state-of-the-art performance. Additionally, it shows competitive results in trajectory and affordance prediction accuracy. These findings validate the effectiveness of the proposed dataset and framework in enhancing robotic brain capabilities. In summary, the main contributions of this paper are as follows:

- We propose ***RoboBrain***, a unified multimodal large language model designed for robotic manipulation, which facilitates more efficient task execution by transforming abstract instruction into concrete actions.
- We meticulously designed the ratio of robotic data to general multi-modal data, implemented a multi-stage training strategy, and incorporated long videos and high-resolution images. This approach provided ***RoboBrain*** with historical frame memory and high-resolution image input, thereby further enhancing its capabilities in robotic manipulation planning.
- We introduce ***ShareRobot***, a high-quality heterogeneous dataset that labels multi-dimensional information, including task planning, object affordance, and end-effector trajectory, effectively enhancing various robotic capabilities.
- Comprehensive experimental results demonstrate that ***RoboBrain*** achieves state-of-the-art performance across various robotic benchmarks, highlighting its potential for real-world applications in robotics.

2. Related Work

MLLM for Robotic Manipulation Planning Existing studies mostly utilize MLLMs primarily focus on understanding natural language and visual observation tasks [6–

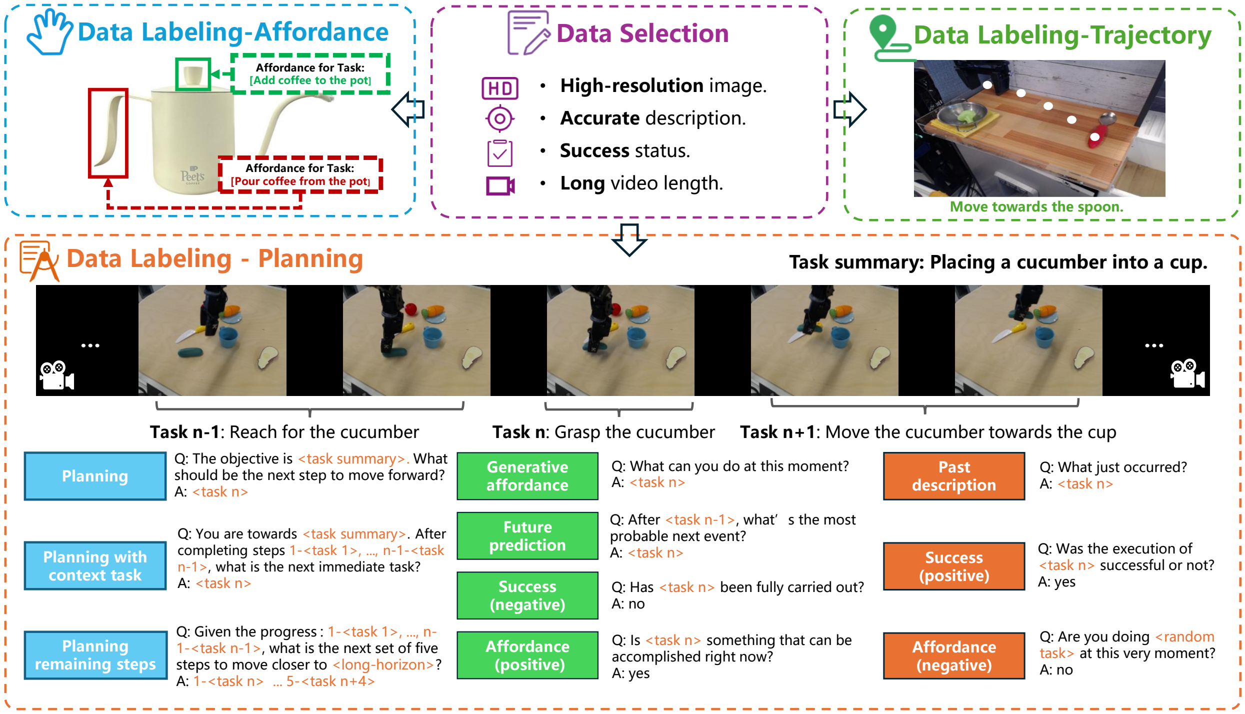


Figure 2. **The generation procession of our ShareRobot dataset.** Our dataset labels multi-dimensional information, including task planning, object affordance, and end-effector trajectories. The task planning is first annotated by atomic tasks and then augmented by constructing question-answer pairs. The affordance and trajectory are labeled on the images according to the specific instructions.

8, 26, 30, 61], with fewer addressing the decomposition of high-level task instructions into actionable steps. PaLM-E [16] generates multimodal inputs by mapping real-world observations into the language embedding space. RT-H [6] and RoboMamba [36] generate reasoning results along with robot actions obtained from an additional policy head. However, while these models generate planning texts and actions, they still lack adequate mechanisms for executing complex atomic tasks, highlighting the need for enhanced affordance perception and trajectory prediction.

Datasets for Manipulation Planning Early datasets for Manipulation [11, 20, 27, 37, 51] mainly comprise annotated images and videos that highlight fundamental hand-object interactions, including grasping and pushing. Recent advancements [15, 21, 50, 52] in robotic manipulation emphasize multi-modal and cross-embodiment datasets for enhanced generalization. Datasets such as RH20T [18], BridgeDataV2 [56], and DROID [25] enhance scene diversity, broadening the range of manipulation scenarios. Notably, RT-X [45] compiles data from 60 datasets across 22 embodiments into the Open X-Embodiment (OXE) repository. In this work, we extract high-quality data from OXE, decompose high-level descriptions into low-level planning instructions, and adapt these into a question-answer format to enhance model training.

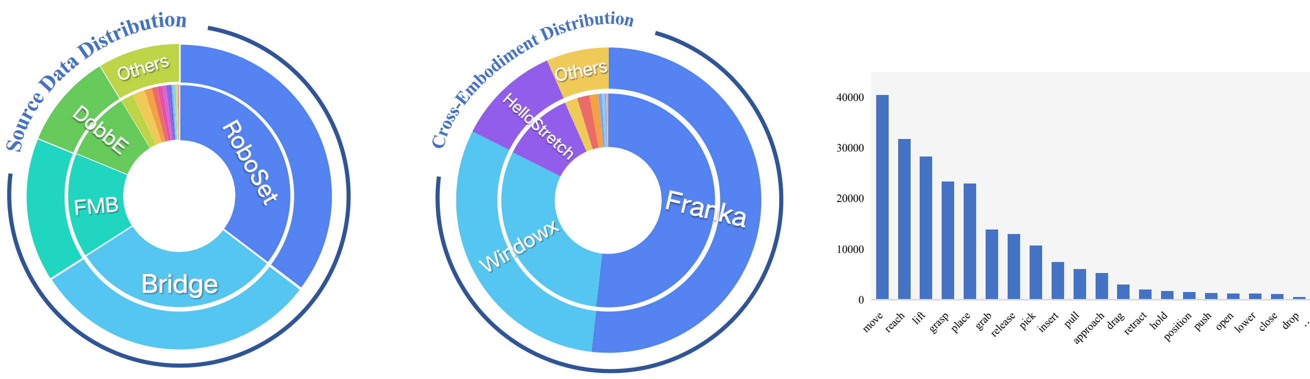
3. ShareRobot Dataset

To enhance the RoboBrain's capability of planning, affordance perception, and trajectory prediction, we develop a dataset called ShareRobot–a large-scale, fine-grained dataset specifically designed for robotic manipulation tasks. The generation procession of our dataset is shown as Fig. 2. The details are described in the following sections.

3.1. Overview

ShareRobot is a comprehensive dataset, facilitates more efficient task execution by transforming abstract concepts into concrete actions. The main features of the ***ShareRobot*** dataset include:

- **Fine-grained** Unlike the Open X-Embodiment dataset [44], which provides generalized high-level task descriptions, each data point in ShareRobot includes detailed low-level planning instructions linked to individual frames. This specificity enhances the model's accuracy in executing tasks at the right moment.
- **Multi-dimensional** To enhance RoboBrain's capabilities from abstract to concrete, we label task planning, object affordances, and end-effector trajectories, allowing for greater flexibility and precision in task processing.
- **High quality** We establish rigorous criteria for selecting data from the Open-X-Embodiment dataset [44], focusing



a. Source Data Distribution

b. Cross-Embodiment Distribution

c. Statics of Types of Atomic Tasks.

Figure 3. **The diversity of our ShareRobot dataset.** Our dataset involves (a) 23 original datasets, (b) 12 embodiments and (c) 107 types of atomic tasks. The distribution of the top 20 most frequent atomic actions within our ShareRobot dataset is presented in (c).

on high resolution, accurate descriptions, successful task execution, visible affordance, and clear motion trajectories. Based on these criteria, we validate 51,403 instances to ensure high quality, forming the foundation for RoboBrain's core capabilities.

- **Large scale** With 1,027,990 question-answer pairs, ShareRobot is the largest open-source dataset for task planning, affordance perception, and trajectory prediction, enabling deeper understanding of complex relationships from abstract to concrete.
- **Rich diversity** In contrast to the RoboVQA [50] dataset's limited scenes, ShareRobot features 102 scenes across 12 embodiments and 107 types of atomic tasks, as shown in Fig. 3. This diversity allows MLLMs to learn from varied real-world contexts, enhancing robustness in complex, multi-step planning.
- **Easy scalability** Our data generation pipeline is designed for high scalability, facilitating expansion as new robotic embodiments, task types, and environments develop. This adaptability ensures the ShareRobot dataset can support increasingly complex manipulation tasks.

3.2. Data Selection

Based on the Open X-embodiment dataset [44], we carefully selected 51,403 instances, mainly focusing on image quality, description accuracy and success status. Our data collection process adheres to the following principles:

- **High-resolution image** We eliminate videos lacking images or those with low resolution. Any video with a resolution below 128 pixels is removed.
- **Accurate description** Videos without descriptions or with vague descriptions are filtered out to avoid affecting the planning capability of the model.
- **Success status** We discard videos of failed tasks, as unsuccessful demonstrations hinder the model's learning.
- **Long video length** Videos with fewer than 30 frames are excluded, as they contain limited atomic tasks.
- **Object not covered** We remove any videos where the target object or end-effector is covered by other objects, as our model has to accurately identify the positions of end-effectors and the object's affordance.
- **Clear Trajectories** We exclude the demonstrations with unclear or incomplete trajectories, as trajectory prediction is one of our RoboBrain's capabilities.

3.3. Data Labeling

Planning Labeling We extract 30 frames from each robotic operation demonstration and use these frames along with their high-level descriptions to decompose them into low-level planning instructions using Gemini [53]. Three annotators then review and refine these instructions to ensure the precision of labeling. Subsequently, we design 5 different templates for each of the 10 question types in RoboVQA [50]. In the process of data generation, we randomly select 2 templates of each question type to generate question-answer pairs for every instance. This process transforms 51,403 instances into 1,027,990 question-answer pairs, with annotators monitoring data generation to maintain the dataset's integrity.

Affordance Labeling We filter 6,522 images and annotate each with affordance areas as $\{l^{(x)}, l^{(y)}, r^{(x)}, r^{(y)}\}$ according to its high-level description, where $\{l^{(x)}, l^{(y)}\}$ are the top left coordinates and $\{r^{(x)}, r^{(y)}\}$ are the bottom right corner coordinates. Subsequently, we conduct a rigorous manual review and refinement of each instruction to ensure its precise alignment with the associated affordance areas.

Trajectory Labeling We filter 6,870 images and annotate each with the gripper's trajectory using at least three $\{x, y\}$ coordinates according to its low-level instruction. Subsequently, we conduct a rigorous manual review and refinement of each instruction to ensure its precise alignment

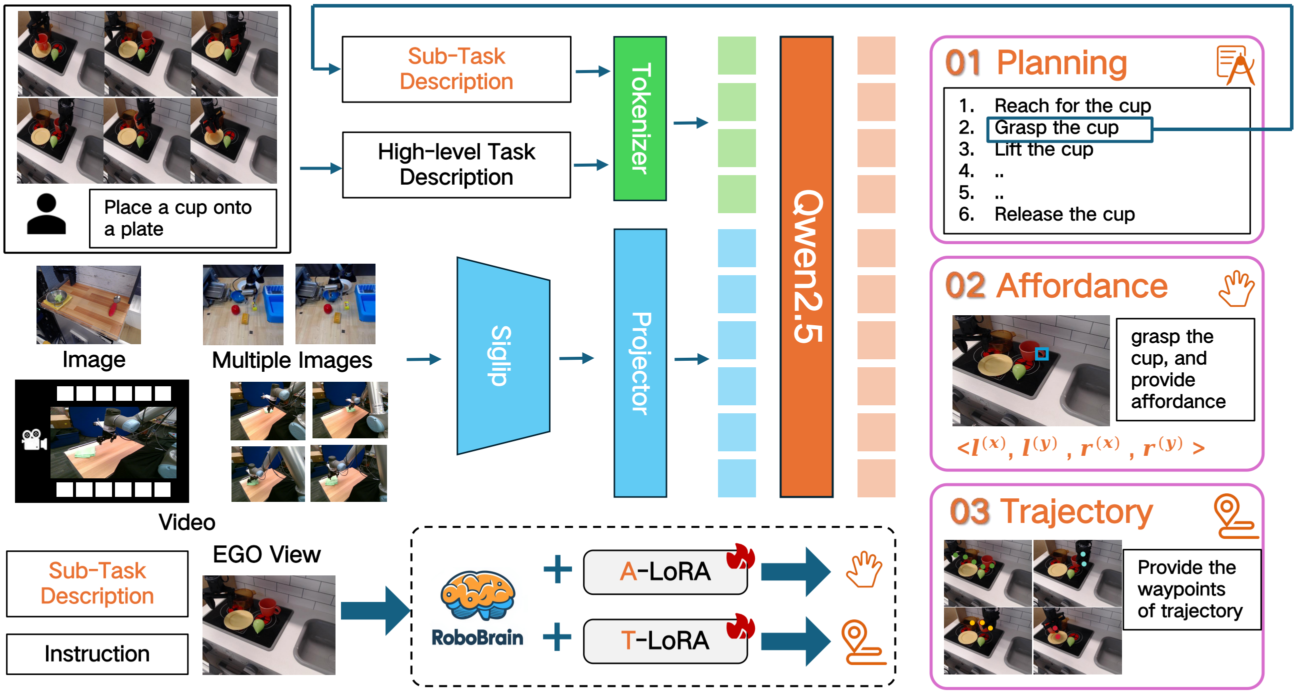


Figure 4. **The pipeline of our RoboBrain.** The images, multiple images, and videos are sent into our model to pre-train a foundation robotic brain. Besides, we fine-tune the RoboBrain via A-LoRA and T-LoRA to develop affordance and trajectory skills. In practical applications, the model first generates detailed plans, and then splits it into sub-task descriptions to execute specific robotic tasks.

with the associated trajectory.

3.4. Data Statistics

We select 23 original datasets from the Open X-embodiment dataset [44]. The distribution of the source data is shown in the Fig. 3. The data involves 102 various scenes (e.g. bedroom, laboratory, kitchen, office), and covers 12 different robot bodies. According to statistics, there are 132 types of atomic actions in this dataset, tasks with higher word frequency are shown in Fig. 3 (c). The 5 most frequent atomic tasks are “pick”, “move”, “reach”, “lift”, and “place”, which are frequent task types in real robotic operation scenarios. This suggests that the distribution of our dataset is reasonable. Finally, we get 1,027,990 question-answer (QA) pairs for planning. For the planning QA pairs dataset, we split 1 million QA pairs as the training set and 2,050 QA pairs as the test set. For the affordance dataset, we split 6,000 images as the training set and 522 images as the test set. For the trajectory dataset, we split 6000 images for training and 870 images for testing.

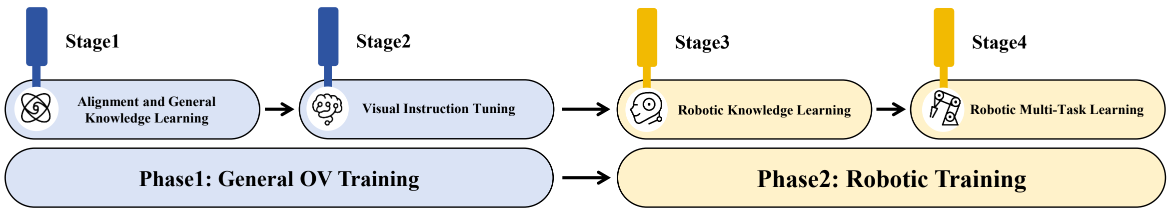
4. RoboBrain Model

In this section, we provide an overview of ***RoboBrain***. Our goal is to enable the Multi-modal Large Language Model (MLLM) to understand abstract instructions and explicitly output object affordance regions and potential operational trajectories, facilitating a transition from abstract to concrete. We employ a multi-stage training strategy: Phase 1 focuses on general OneVision (OV) training to develop a foundational MLLM with strong understanding and instruction-following abilities. Phase 2, the robotic training phase, aims to empower the core capabilities of RoboBrain from abstract to concrete.

4.1. Model Architecture

RoboBrain consists of three modules: the foundational model for planning, the A-LoRA model for affordance perception, and the T-LoRA model for trajectory prediction. In practical applications, the model first generates detailed plans, and then splits it into sub-task descriptions to execute affordance perception and trajectory prediction. The pipeline of our RoboBrain is shown to Fig. 4.

Foundational Model for Planning We utilize LLaVA as the foundational model for RoboBrain, which consists of three main modules: the Vision Encoder (ViT) $g(\cdot)$, the Projector $h(\cdot)$, and the Large Language Model (LLM) $f(\cdot)$. Specifically, we employ SiglIP [59], a 2-layer MLP [33], and Qwen2.5-7B-Instruct [54]. Given an image or video X_v as visual input, ViT encodes it into visual features $Z_v = g(X_v)$, which are then mapped to the semantic space of the LLM through Projector, resulting in a sequence of visual tokens $H_v = h(Z_v)$. Finally, the LLM generates a textual



		Stage-1	Stage-1.5	Stage-2		Stage-3	Stage-4	
				Single-Image	OneVision		A-LoRA	T-LoRA
Vision	**Resolution**	384	Max 384×{2×2}	Max 384×{6×6}	Max 384×{6×6}	Max 384×{6×6}	Max 384×{6×6}	Max 384×{6×6}
	#Tokens	729	Max 729×5	Max 729×37	Max 729×37	Max 729×37	Max 729×37	Max 729×37
Data	**Dataset**	LCS	Image	Image	Image & Video	Robotic Data	Afford. Data	Traj. Data
	#Samples	558K	4M	3.2M	1.6M	3M	10K	400K
Model	**Trainable**	Projector	Full Model	Full Model	Full Model	Full Model	A-LoRA	T-LoRA
	#Tunable Parameters	17.0M	8.0B	8.0B	8.0B	8.0B	28.0M	28.0M
Training	**Batch Size**	8	2	1	1	1	4	4
	LR: ψ_{ViT}	-	2×10^{-6}	2×10^{-6}	2×10^{-6}	2×10^{-6}	2×10^{-6}	2×10^{-6}
	LR: $\{\theta_{\text{Proj.}}, \phi_{\text{LLM}}, \phi_{\text{LoRA}}\}$	1×10^{-3}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}
	Epoch	1	1	1	1	1	1	1

Table 1. Detailed configuration for each training stage of the RoboBrain.

response in an autoregressive manner based on the human language instruction X_t and H_v.

A-LoRA Module for Affordance Perception The term affordance in our work refers to the area where the human hand makes contact with objects. During interactions, humans instinctively engage with various objects within specific regions. We utilize *bounding boxes* to represent affordances. Formally, consider an image I consisting of multiple objects with their affordances: $O_i = \{A_i^0, A_i^1, ..., A_i^N\}$, where the ith object owns N affordances. The format of affordance is defined as $\{l^{(x)}, l^{(y)}, r^{(x)}, r^{(y)}\}$, and $\{l^{(x)}, l^{(y)}\}$ represents the top left corner coordinates of affordance, while $\{r^{(x)}, r^{(y)}\}$ is the bottom right corner coordinates.

T-LoRA Module for Trajectory Prediction The term trajectory in our work refers to the concept of *2D visual traces*, as presented in [19]. We define trajectory waypoints as a series of 2D coordinates representing the movement of the end-effector or hand throughout the process. Formally, at time step t, the trajectory waypoints can be represented as $P_{t:N} = \{(x_i, y_i) \mid i = t, t+1, \ldots, N\}$, where (x_i, y_i) denotes the i-th coordinate in the visual trace, and N represents the total number of time steps in the episode.

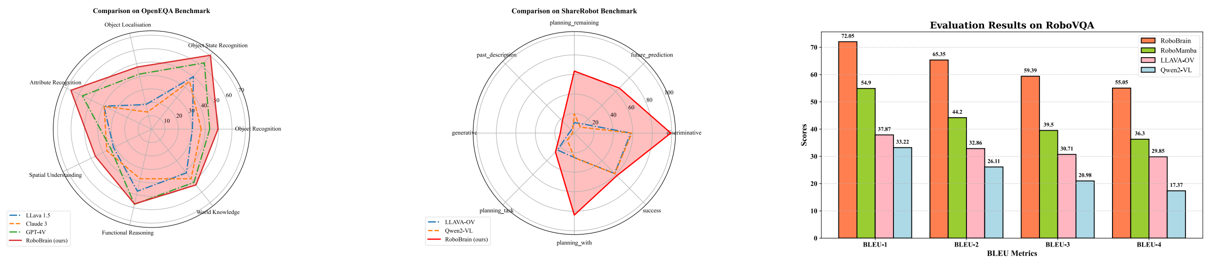
4.2. Training

Phase 1: General OV Training In Phase 1, we drew on the state-of-the-art training data and strategies from LLaVA-OneVision [28] to construct a foundational model with general multi-modal understanding and visual instruction following capabilities. This lays the groundwork for enhancing the model's robotic manipulation planning abilities in Phase 2. Detailed information is provided in Tab. 1.

In Stage 1, we utilize the image-text data from the LCS-558K dataset [10, 49] to train Projector, facilitating the alignment of visual features Z_v with the LLM semantic features H_v. In Stage 1.5, we train the entire model using 4M high-quality image-text data to enhance the model's multi-modal general knowledge understanding capabilities. In Stage 2, we further train the entire model with 3.2M single-image data and 1.6M image and video data from LLaVA-OneVision-Data [28], aiming to enhance the instruction-following abilities of RoboBrain and improve understanding of high-resolution image and video.

Phase 2: Robotic Training In Phase 2, we build upon the robust multi-modal foundational model developed in Phase 1 to create a more powerful model for robotic manipulation planning. Specifically, we aim for RoboBrain to understand complex, abstract instructions, support the perception of historical frame information and high-resolution images, and output object affordance regions while predicting potential manipulation trajectories. This will facilitate the transition from abstract to concrete in manipulation planning tasks. Detailed information is provided in Tab. 1.

In Stage 3, we collected a dataset of 1.3M robotic data to improve the model's manipulation planning capabilities. Specifically, this data is sourced from RoboVQA-800K [50], *ScanView-318K* including MMScan-224K [23, 40], 3RScan-43K [23, 55], ScanQA-25K [4, 23], SQA3d-26K [23, 41], and a subset of ShareRobot-200K introduced in this paper. These datasets contain substantial scene-scanning image data, long video data, and high-resolution



(a) OpenEQA Benchmark (b) ShareRobot Benchmark (c) RoboVQA Benchmark

Figure 5. The performance of our model RoboBrain on the OpenEQA, ShareRobot, and RoboVQA benchmarks. RoboBrain surpassed all baseline models, achieving state-of-the-art results.

data to support the model's ability to perceive diverse environments. Additionally, the fine-grained, high-quality planning data in the ShareRobot dataset enhances the manipulation planning capabilities of RoboBrain. To mitigate the issue of catastrophic forgetting [60], we selected a high-quality subset of approximately 1.7M image-text data from Phase 1 to mix with the robotic data collected in Stage 3 for training, tuning the entire model accordingly. In Stage 4, we enhanced the model's ability to perceive object affordances and predict manipulation trajectories from instructions, utilizing affordance and trajectory data from the ShareRobot dataset and other open-source sources [39, 43]. This was achieved by incorporating LoRA modules during training for concrete manipulation capabilities.

5. Experiment

5.1. Implementation Details

During the entire training phase, we employed the Zero3 [48] distributed training strategy, conducting all experiments on a cluster of servers, each equipped with 8×A800 GPUs. The training components for each stage, including image resolution settings, batch size, epochs, and learning rates, are provided in Tab. 1.

5.2. Evaluation Metrics

Planning Task We selected RoboVQA [50], OpenEQA [42], and the test set of ShareRobot as robotic benchmarks for multi-dimensional assessment. For RoboVQA, we adopt the BLEU1 to BLEU4 metrics [47] used in RoboMamba [36] for evaluation. Additionally, for OpenEQA and ShareRobot, we use GPT-4o [46] as the evaluation tool, scoring based on the alignment or similarity between model predictions and ground truth, which serves as the final performance score for the model.

Affordance Prediction We utilize the Average Precision (AP) to evaluate the affordance performance of our model. AP metric summarizes the precision-recall affordance curve, which plots the relationship between precision and recall at various threshold settings. It is calculated across multiple Intersection over Union (IoU) thresholds to obtain a more comprehensive evaluation.

Trajectory Prediction We evaluate the similarity between ground truth and predicted trajectories, both represented as sequences of 2D waypoints normalized to $[0, 1000)$, following Qwen2-VL [58]. The evaluation uses three metrics: Discrete Fréchet Distance (DFD) [19], Hausdorff Distance (HD), and Root Mean Square Error (RMSE). DFD captures overall shape and temporal alignment, HD identifies maximum deviation, and RMSE measures average pointwise error. Together, these metrics provide a comprehensive assessment of trajectory accuracy and similarity.

5.3. Evaluation on Robot Brain Task

Evaluation on Planning Task We selected 6 powerful MLLMs as our baselines for comparison, including both open-source and closed-source models with different architectures. Specifically, these models include GPT-4V [2], Claude3 [1], LLaVA-1.5 [34], LLaVA-OneVision-7b [28], Qwen2-VL-7b [57] and RoboMamba [36]. Our specific experimental results are shown in Fig. 5. Our RoboBrain outperformed all baseline models across three robotic benchmarks. RoboBrain significantly outperformed all baseline models on OpenEQA and ShareRobot, which can be attributed to its robust capabilities in understanding robotic tasks and perceiving long videos. Additionally, this pattern was observed in other benchmarks as well, with RoboBrain consistently demonstrating superior performance on RoboVQA, achieving a BLEU-4 score that exceeded that of the second-place model by 18.75. This result highlights its capability to decompose complex long-range task planning.

Evaluation on Affordance Prediction Our results are summarized in Tab. 2. We compare the Qwen2-VL-7B and LLaVA-NeXT-7B models. Qwen2-VL [57] has a superior visual grounding ability and LLaVA-NeXT [31] owns a high-resolution and strong vision tower. We test them all on the AGD20K affordance test set. Our RoboBrain outperforms significantly the other models. It surpasses Qwen2-

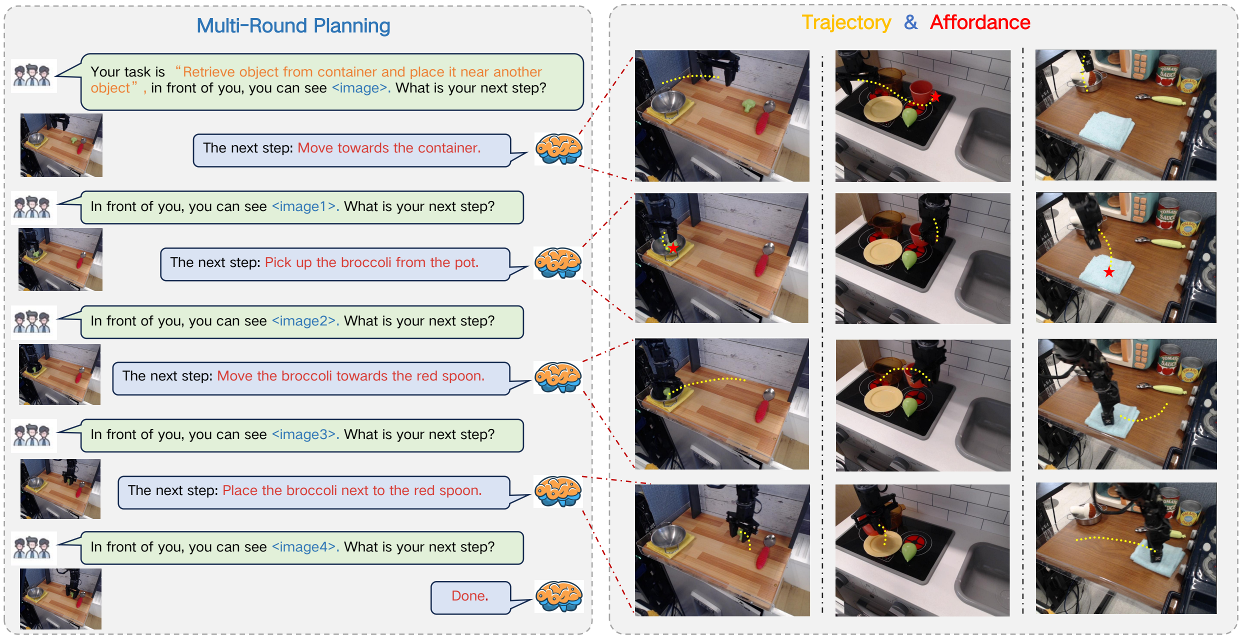


Figure 6. This visualization illustrates that RoboBrain can interpret human instructions and visual images to generate action plans and assessments based on real-time image feedback. Furthermore, it predicts trajectories for each step and identifies corresponding affordances.

Table 2. **The comparison of affordance prediction.** We utilize AP as the metric, and test them on affordance test set.

Model	AP ↑
LLaVA-NeXT-7B [34]	9.8 %
Qwen2-VL-7B [5]	12.5 %
RoboBrain (Ours)	**27.1** % (14.6↑)

VL [57] by 14.6 AP, and LLaVA-NeXT by 17.3 AP. It validates our RoboBrain can understand the physical properties of objects and provide the affordance accurately.

Evaluation on Trajectory Prediction We compare several variants of our model, and the results are in Tab. 3: **(1) Baseline**, fine-tuned on trajectory-related VQA data; **(2) Start_Points**, which adds the 2D start coordinates of the end-effector; **(3) Max_Points**, limiting waypoints to 10 via uniform sampling; and **(4) Spec_Token & End_Points**, which adds end-effector positions and special tokens to emphasize waypoints and start/goal points. Each variant builds on the previous one, with the final model integrating all components. Our most effective model integrates all design choices. As shown in the last row of Tab. 3, DFD, HD, and RMSE decreased by 42.9%, 94.2%, and 31.6%, respectively, compared to the baseline. We found that adding start points corrected the translational offset between the generated trajectory and the end-effector.

Table 3. **Trajectory Prediction Results Comparison.** Discrete Fréchet Distance (DFD), Hausdorff Distance (HD), and Root Mean Square Error (RMSE).

Method	DFD ↓	HD ↓	RMSE ↓
RoboBrain (Base)	0.191	0.171	0.133
+ Start_Points	0.176	0.157	0.117
+ Max_Points	0.185	0.163	0.125
+ Spec_Token	**0.109** (42.9%↓)	**0.010** (94.2%↓)	**0.091** (31.6%↓)

5.4. Visualization

In this section, we present visual examples of RoboBrain in Fig. 6. Given human instructions and visual inputs, RoboBrain engages in multi-turn interactions, understanding and planning future steps. It also outputs more concrete affordances and trajectories.

6. Conclusion

In this paper, we introduce ***ShareRobot***, a high-quality dataset that labels multi-dimensional information, including task planning, object affordance, and end-effector trajectory. We also present ***RoboBrain***, an MLLM-based model that integrates robotic and general multi-modal data, employs a multi-stage training strategy, and leverages long videos and high-resolution images to enhance robotic manipulation. Extensive experiments demonstrate that RoboBrain achieves state-of-the-art performance across various robotic tasks, underscoring its potential to significantly advance robotic capabilities.

Acknowledgments

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